Four Channel Groups from One Cost

A Machine-Verified Decomposition of the Recognition Science Death Projection

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March 2026

Abstract

The Recognition Composition Law (RCL) forces an 8-channel octave structure whose death projection decomposes into exactly four groups with multiplicities $(3, 1, 3, 1)$. Two structural features are forced: the octave midpoint partitions channels into low-frequency (< 4) and persistent (≥ 4) halves, and the maximum-index slot uniquely maximizes the preservation budget. Two bridge hypotheses complete the decomposition: (H1) the transition boundary sits at the midpoint minus one, giving a 3+1+4 split; (H2) the transitional gate opens at the first reflexivity level where cost exceeds kernel dimension, forcing k=3. Under H1–H2, the resulting four groups have distinct survival laws and support a unique rank-preserving bijection to any ordered four-level external taxonomy. All results are formalized in Lean 4 with zero `sorry` obligations. Every claim is labeled as forced, bridge, or interpretive.

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1 Introduction

This paper studies the internal structure of the Recognition Science (RS) death projection. The central question is: *what forced decomposition does the 8-channel octave admit when an embodied recognition boundary dissolves?*

The answer is partly forced and partly bridged. The RS core already determines an 8-channel octave, a monotone preservation budget $B(k) = \varphi^k$, and a unique maximum-budget slot. Two additional bridge hypotheses complete a four-group decomposition with multiplicities $(3, 1, 3, 1)$.

To maintain rigorous provenance, every claim carries one of three labels:

- **Forced**: derived from the RS core (T0–T8) and machine-verified in Lean 4.
- **Bridge hypothesis**: a principled structural claim not yet derived from the RS core.
- **Interpretive**: a semantic identification connecting the formal structure to external language.

1.1 Scope

This is a mathematical investigation within the RS axiom system. It derives what the axioms force under stated hypotheses. It is not an empirical paper and does not attempt to establish any historical, phenomenological, or theological claim. The derivation is machine-verified in Lean 4 with zero `sorry` obligations.

2 Related Work

Several mathematical frameworks model aspects of consciousness. Tononi's Integrated Information Theory (IIT) [1, 2] assigns an integration scalar to a system, Penrose's work [3] links consciousness to foundational physics, and Hoffman's conscious realism [4] treats conscious agents as primitive. None of these derive a discrete four-group decomposition of a death transition from a single functional equation. In that sense, the present paper is not a survey of consciousness models but a provenance analysis of one specific RS claim.

3 The RS Forcing Chain

Recognition Science derives structure from the **Recognition Composition Law** (RCL):

$$J(xy) + J(x/y) = 2J(x)J(y) + 2J(x) + 2J(y). \tag{1}$$

With normalization $J(1) = 0$ and calibration $J''_{\ln}(0) = 1$, the unique solution is [6]

$$J(x) = \tfrac{1}{2}(x + x^{-1}) - 1 = \cosh(\ln x) - 1. \tag{2}$$

The complete forcing chain (T0–T8) derives, in order: logic, the meta-principle, discreteness, ledger structure, recognition events, the unique cost, the golden ratio, the 8-tick period, and spatial dimension $D = 3$.

For the present paper, the critical forced outputs are:

(i) φ is the unique positive root of $x^2 = x + 1$. **Status: Forced.** Lean: `phi_sq_eq`.

(ii) The 8-tick period yields 8 abstract information channels. **Status: Forced.** Lean: `octave_slot_count_from_t7`.

(iii) The reflexivity eigenvalue satisfies $\lambda^2 = \lambda + 1$, hence $\lambda = \varphi$. **Status: Forced.** Lean: `reflexivityEigenvalue_sq`.

(iv) The capacity recursion $C(0) = 1$, $C(k{+}1) = \varphi \cdot C(k)$ gives $B(k) = \varphi^k$. **Status: Forced.** Lean: `informationCapacity_eq_pow`.

4 The Death Projection

When an embodied recognition boundary dissolves, RS models the transition as a diagonal projection $\mathcal{D} : \mathcal{H}_{\text{emb}} \to \mathcal{H}_{\text{light}}$ on an 8-channel state.

Two facts are forced from the RS core:

Theorem 4.1 (Z-Pattern Conservation)**.** *The identity invariant Z is preserved through dissolution and re-embodiment.*
Status: Forced. *Lean: `dissolution_certificate_preserves_Z`, `Z_survives_death`, `Z_survives_rebirth`.*

Theorem 4.2 (Projection Monotonicity)**.** *For any admissible retention mask $0 \le m_j \le 1$, $I_{\text{proj}} \le I_{\text{tot}}$.*
Status: Forced. *Lean: `projectedInformation_le_total`.*

The unresolved question is the internal grouping of these channels under dissolution.

5 Forced Structural Features

Two features of the 8-channel death projection are forced without additional hypotheses.

5.1 Octave Midpoint

Theorem 5.1 (Octave Midpoint)**.** *The 8-tick octave has a structural midpoint $N_{\text{mid}} = \lfloor 8/2 \rfloor = 4$. This divides the 8 channels into:*

- *a low-frequency half: slots $0, 1, 2, 3$,*
- *a persistent half: slots $4, 5, 6, 7$.*

Status: Forced. *Lean: `octaveMidpoint_from_eight_tick`.*

The persistent half matches the proved condition $\text{isZStructural}(c) \iff \iota(c) \ge 4$ (Lean: `z_structural_iff_above_midpoint`).

5.2 Maximum-Budget Slot

Theorem 5.2 (Budget Maximizer)**.** *The preservation budget $B(k) = \varphi^k$ is strictly monotone. Among the 8 channels, slot 7 uniquely maximizes the budget:*

$$B(\iota(c)) < B(7) \quad \forall c \neq \text{reflexivity}.$$

Status: Forced. *Lean: `reflexivity_is_max_slot`, `z_carrier_at_top`.*

This is purely structural: φ^k increases with k, and the slot index never exceeds 7.

6 Bridge Hypotheses

Two bridge hypotheses, not yet derived from the RS core, are needed to complete the four-group decomposition.

Hypothesis 6.1 (Transition Boundary at $N_{\text{mid}} - 1$)**.** *The boundary between substrate-dependent and transitional channels sits at $N_{\text{mid}} - 1 = 3$, not at $N_{\text{mid}} = 4$. That is:*

$$\text{substrate-dependent}(c) \iff \iota(c) < 3, \tag{3}$$

$$\text{transitional}(c) \iff \iota(c) = 3. \tag{4}$$

Status: Bridge hypothesis. *A derivation would need to show that the DFT-8 conjugate pairing gives slot 3 a distinguished transitional role. The current formalization proves only consistency with the existing bridge predicates.*

Hypothesis 6.2 (Cost-Crossing Transitional Gate)**.** *The transitional channel activates at the first reflexivity level k where $\varphi^k - 1 \geq \dim(\ker \mathcal{D})$.*
Status: Bridge hypothesis. *The numerical consequence is clean, but the principle that the gate criterion is cost-to-kernel crossing is not currently derived from the RS axioms.*

Theorem 6.3 (Threshold under H6.2)**.** *Under Hypothesis 6.2, the transitional threshold is $k^* = 3$.*

Proof. $\varphi^2 - 1 = \varphi < 3$ while $\varphi^3 - 1 > 3$. **Status: Forced.** Lean: `emotional_threshold_forced_at_three`. □

7 The Four-Group Decomposition

Under Hypotheses 6.1 and 6.2, the 8 channels decompose into four groups with distinct post-dissolution behavior.

Theorem 7.1 (Four-Group Decomposition under H1–H2)**.** *The channel-group assignment is determined by the octave-slot index:*

$$\text{group}(c) = \begin{cases} \text{Group}_0 & \iota(c) < 3, \\ \text{Group}_1 & \iota(c) = 3, \\ \text{Group}_3 & \iota(c) = 7, \\ \text{Group}_2 & 4 \leq \iota(c) \leq 6. \end{cases}$$

The four groups have cardinalities $(3, 1, 3, 1)$ summing to 8.
Status: Forced. *Conditional on H1–H2. Lean:* `channelToGroup_forced_by_structure`, `four_classes_forced`.

Proof. Hypothesis 6.1 separates slots < 3 from slot 3. Theorem 5.2 separates slot 7 from slots 4–6. The remaining multiplicities follow by enumeration. □

Group	Channels	Survival law	Status
Group_0	0–2	zero survival	conditional on H1
Group_1	3	threshold-gated survival	conditional on H1+H2
Group_2	4–6	finite persistent budget	forced from midpoint split
Group_3	7	maximal budget, Z-carrier candidate	forced from max-budget

7.1 Maximum-Budget Identification

We identify the conserved Z-pattern with the maximum-budget slot (slot 7). This is a natural structural identification, motivated by the principle that an indestructible invariant should occupy the most persistent channel.
Status: Bridge hypothesis. Deriving the principle "conserved topological invariants maximize budget" from the RS core remains an open problem.

8 Ordered Four-Level Interfaces

The four-group decomposition determines a generic interface theorem.

Definition 8.1 (Dissolution Rank)**.** Let $r : \text{ChannelGroup} \to \text{Fin}(4)$ be the injective rank function

$$r(\text{Group}_0) = 0, \quad r(\text{Group}_1) = 1, \quad r(\text{Group}_2) = 2, \quad r(\text{Group}_3) = 3.$$

Theorem 8.2 (Rank-Preserving Uniqueness)**.** *Let α be any four-element set with a rank $\rho : \alpha \to \text{Fin}(4)$. Any bijection $f : \alpha \to \text{ChannelGroup}$ satisfying $r(f(a)) = \rho(a)$ for all a is unique.*
Status: Forced. *Lean:* `rank_preserving_map_unique`.

Proof. If f and g both preserve rank, then $r(f(a)) = \rho(a) = r(g(a))$ for each a. Injectivity of r implies $f(a) = g(a)$. □

Corollary 8.3 (Universal Four-Level Interface)**.** *Any ordered four-level external taxonomy admits exactly one rank-preserving bijection to the RS channel groups.*

Remark 8.4. The current codebase contains several concrete four-level instantiations of this interface theorem. The present paper suppresses their historical or external naming and records only the generic structural result.

9 Dissolution Timing

The φ-ladder yields a strict hierarchy of persistence:

Theorem 9.1 (Dissolution Hierarchy)**.**

$$r(\text{Group}_0) < r(\text{Group}_1) < r(\text{Group}_2) < r(\text{Group}_3).$$

Status: Forced. *Lean:* `dissolution_strict_hierarchy`.

9.1 Structural Ordering Predictions

Without a calibrated base timescale, RS predicts only the ordering of dissolution:

1. Group_0 dissolves first.
2. Group_1 dissolves next if activated.
3. Group_2 persists longer with a finite budget.
4. Group_3 is maximally persistent.

9.2 Budget Bounds

The relative scale of persistence is governed by:

Group	Budget	Numerical range
Group_0	0	zero survival
Group_1	φ^3 (conditional)	$4.0 < B < 4.25$
Group_2	$[\varphi^4, \varphi^6]$	$6.9 < B < 17.9$
Group_3	φ^7	$B > 29$

Quantitative time calibration remains open: a physical base period must be specified before the ladder values can be converted into durations.

10 Formal Verification

The derivation is formalized in a single Lean 4 module.

Metric	Value
Lines	658
Definitions	22
Theorems	40+
`sorry`	0
Build time	∼12 s

Key formal results include:

Result	Lean identifier
Midpoint split	`partition_forced_by_octave`
Threshold $k = 3$ under H2	`emotional_threshold_forced_at_three`
Maximum-budget slot	`max_slot_is_reflexivity`
Group structure under H1–H2	`channelToGroup_forced_by_structure`
Multiplicity $(3, 1, 3, 1)$	`four_classes_forced`
Generic interface uniqueness	`rank_preserving_map_unique`
Master certificate	`steiner_four_bodies_forced_certificate`

11 Discussion

11.1 What Is Forced

- The 8-channel octave.
- The midpoint at 4.
- The monotone budget $B(k) = \varphi^k$.
- The maximum-budget slot at index 7.
- The generic uniqueness of any rank-preserving four-level interface.

11.2 What Remains Bridged

- The boundary at $N_{\text{mid}} - 1$ rather than N_{mid}.
- The cost-crossing gate principle.
- The identification of Z with the maximum-budget slot.

- Any semantic interpretation assigned to the groups.
- Any physical calibration of the dissolution timescale.

11.3 Open Problems

1. Derive the transition boundary from DFT-8 conjugate pairing.
2. Derive the gate principle from an $\hat{R}$-optimality theorem.
3. Derive the invariant-maximizes-budget principle.
4. Calibrate the ladder values in physical time units.
5. Derive semantic channel labels from the RS core.

12 Conclusion

The Recognition Composition Law forces a substantial part of the death-projection architecture: an 8-channel octave, a midpoint split, a monotone preservation budget, a unique maximum-budget slot, and a generic uniqueness theorem for ordered four-level interfaces. Two bridge hypotheses then complete a four-group decomposition with multiplicities $(3, 1, 3, 1)$.

The result is therefore not yet a full inevitability theorem, but it is already a sharply constrained structural theorem: the RS death projection can support only one ordered four-level interface once the two bridge choices are fixed. Closing those two bridge gaps would upgrade the decomposition from a principled structural consequence to an unavoidable theorem of the Recognition Composition Law.

References

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[3] R. Penrose, *Shadows of the Mind*, Oxford: Oxford University Press, 1994.

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[5] J. Washburn, "The Death Transition in Recognition Science: Provenance Draft of the Fredholm Program," Recognition Science Technical Report, 2025.

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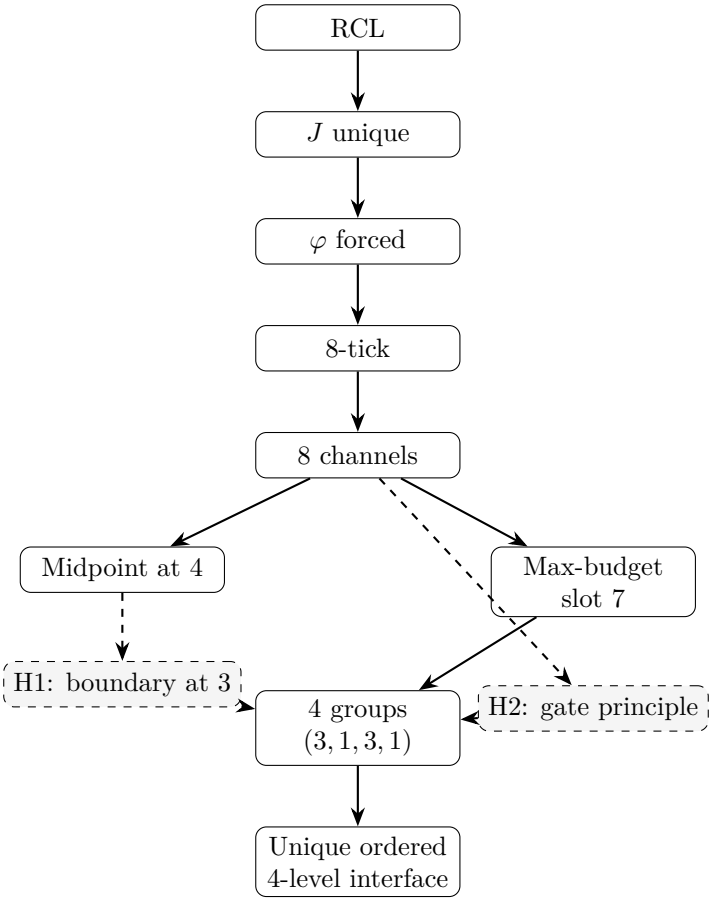
A Forcing Diagram



Solid boxes: forced from the RS core. Dashed boxes: bridge hypotheses.